TECHNICAL COMPETENCY AND DATA INTEGRITY

01

WHAT IS DIGITAL LITERACY?

The ability to discover, evaluate, and communicate technical information through digital platforms.

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- Data Fluency: Interpreting complex datasets and simulation results.
- Technical Synthesis: Combining information from CAD models, research papers, and AI tools.
- Critical Thinking: Evaluating the accuracy of algorithmic outputs and automated calculations.

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- Lifelong Technical Adaptability: A "growth mindset" focused on continually upskilling as new industry-standard software and hardware emerge during your career.
- Ethical Digital Citizenship: Understanding the legal and ethical implications of using copyrighted technical materials, avoiding plagiarism, and respecting intellectual property.

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- Digital Content Synthesis: The skill of creating professional-grade technical artifacts, such as CAD models, 3D prototypes, and multimedia reports that effectively communicate engineering concepts.
- Critical Evaluation of AI & Algorithms: The ability to question and verify the outputs of automated tools and AI models to ensure structural integrity and mathematical accuracy.

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THE ENGINEER`S DIGITAL TOOLKIT

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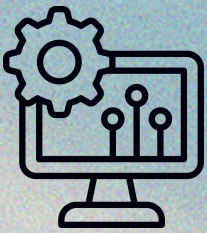
- Design & Modeling (CAD):
  - Tools: AutoCAD, SolidWorks, or Fusion 360.
  - Purpose: Creating 2D drafts and 3D models for mechanical parts, structural designs, or circuit layouts.



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- Computation & Data Analysis:
- Tools: MATLAB, Python (NumPy/Pandas), or Excel (Advanced).
- Purpose: Automating complex mathematical calculations, processing sensor data, and creating technical graphs.

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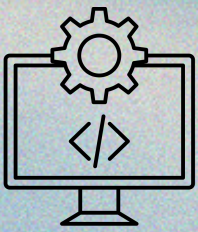
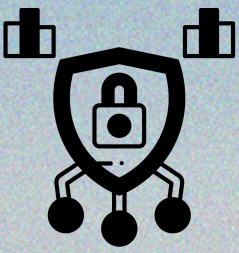
- Real-Time Collaboration:
- Tools: Google Workspace (Docs/Sheets) or Microsoft Teams.
- Purpose: Collaborative drafting of lab reports and version control for group design projects.





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- Visual Communication:
- Tools: Canva.
- Purpose: Designing professional project posters, flowcharts, and stakeholder presentations to simplify technical data.

TECHNICAL COMPETENCY AND DATA INTEGRITY

SAFE & ETHICAL INTERNET PRACTICES

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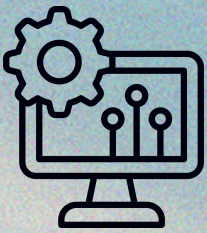
- Data Integrity & Verification:
- Action: Always verify AI-generated formulas, code, or technical data against peer-reviewed textbooks or established engineering standards (e.g., ISO, IEEE).
- Ethics: Automated tools can produce "hallucinations" or biased results; the engineer is ultimately responsible for the safety of the final design.



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- Intellectual Property (IP) Respect:
- Action: Use tools like Zotero or Mendeley to properly cite research papers, CAD models, and open-source code.
- Ethics: Avoid "ethics-washing" or plagiarism; respecting patents and copyrights is essential for professional credibility and legal compliance.

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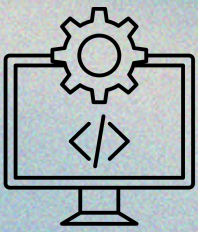
- Robust Cyber Hygiene:
- Action: Use Multi-Factor Authentication (MFA) for all academic and lab accounts and a reputable Password Manager to secure proprietary project data.
- Safety: Protect sensitive simulation data and experimental results from unauthorized access or breaches.



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- Responsible Innovation:
- Action: Evaluate the social and environmental impact of the digital tools and technologies you implement in your projects.
- Ethics: Strive for "Secure by Design" principles to mitigate potential harm to society from the outset of the development process.